



DAVID LU

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EXPERIENCE

Senior Software Engineer at Hedra - 2025 to 2026

- Core engineer building system for multi-step agentic video creation
 - Directed and architected agent harness in Pydantic AI
 - Integrated AI features from FAL, Together, Anthropic, OpenAI and ElevenLabs
 - Capable of generating short viral videos and podcasts from scratch
- Built web-native video player and timeline editor with feature-parity to Canva
- Worked within a small team to completely overhaul Hedra platform from scratch
 - App launch doubled revenue and tripled user base for Hedra

Concept Artist at Nickelodeon - 2022 to 2024

- Designed props, backgrounds, characters and color keys for the Plankton Movie, Transformers EarthSpark and unannounced series

Software Engineer at Google – 2016 to 2022

- Responsible for building web app frontends and games on Google homepage (Doodles)
 - Hundreds of million of plays and centuries of playtime on aggregate
- Architect of Entity-Component-System game engine used in most large doodles
 - Directed team of artists and engineers in ECS development
- Designed and ported mobile native experiences for Google iOS App and Google Android
- Lead engineer and Game Designer for 2021 Doodle Champion Island Game
 - Primary designer for minigames, questing, dialogue, achievement and physics
 - Largest doodle in history in terms of scope, playtime and games played per day
- Lead Engineer on 2017 Hip Hop Doodle featuring DJ Turntable audio editor app
 - Implemented vinyl scratching interface, beat-matching functionality, multichannel crossfaders and mobile/chromeOS multitouch
- Project Lead and designer for 2021 Pizza Doodle and 2020 Wackeldackel Doodle

EDUCATION

University of Toronto at St. George, Computer Science – 2012 to 2015

- Honours Bachelor of Science, specializing in Machine Learning

SKILLS

Language: Dart, Java, Objective-C, Python, SQL, Swift, TypeScript

Frontend: CreateJS, Flutter, Next.js, React Native, React, Unity

Backend: AWS, Docker, FastAPI, Firebase, Flask, Kubernetes, Node.js, PostGRES

AI: Diffusers, LangGraph, Pydantic AI, PyTorch (einops, np, torchvision), TensorFlow